

General Instructions

We would like to better understand your data engineering mindset and thought process.

Please submit the description of the execution plan, the code, the output, and any comments.

You can use any tool/programming language/ library you wish. Please include any dependencies/ instructions required to recreate the output.

Data

The dataset that you will work with is located at

<https://www.kaggle.com/usdot/flight-delays?select=flights.csv>

It contains flight data regarding 2015 US flights. Each row can be identified by

(YEAR, MONTH, DAY, AIRLINE, FLIGHT_NUMBER, TAIL_NUMBER, SCHEDULED_DEPARTURE)

Tasks

Task 1: We would like to store this dataset ('flights') and update it in a daily basis. We can get a daily feed ('feed') that contain information regarding the rolling last n flight days. We know that the 'feed' may contain updated/corrected flight details regarding the 7 most recent days. Using the available data, please describe your selection of n, how you would model this flow, which tools you would use, and any sanity checks that you would set.

Task 2: Using 'flights', we would like to create a new dataset ('flights_availability') that gets updated daily and holds information regarding the ground time, i.e., the total time that each aircraft was not flying in each flight day. You can use columns DEPARTURE_TIME, ARRIVAL_TIME, ELAPSED_TIME to identify time of flight.

Please create two additional columns, *availability*, and *night_availability*

Regarding *night_availability*, for each flight day, we want to get the total ground time available between 23:00 and 07:00

The final dataset 'flights_availability' should contain the aggregate view on the airline level, for each flight day, each airline, total *availability* and total *night_availability*.

Thank you